

NAME : SHAH FAISAL

ROLL NO : 23I-0058

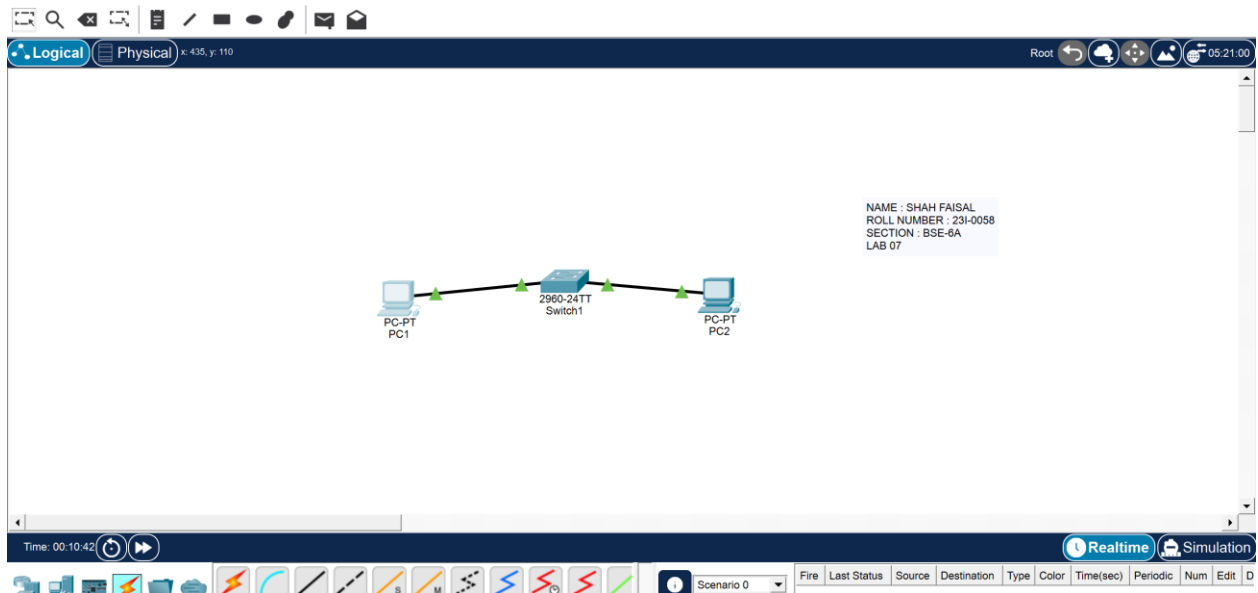
SECTION : BSE – 6A

Computer Networks Lab

BSSE SPRING 2026

LAB 07

Task 1: MAC Address Table and Switching (5 Marks)



PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address:

Link Local Address: FE80::2D0:58FF:FECD:8712

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

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Switch1

Physical
 Config
 CLI
 Attributes

IOS Command Line Interface

```

* 1 26 WS-C2960-24TT-L 15.0(2)SE4 C2960-LANBASEK9-M

Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnguyen

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>enable
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
Switch#

```

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PC1

Physical
 Config
 Desktop
 Programming
 Attributes

Command Prompt

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

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Switch1

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>enable

Switch#show mac address-table

Mac Address Table

Vlan	Mac Address	Type	Ports
1	00d0.58cd.8712	DYNAMIC	Fa0/1
1	00d0.bc71.65c7	DYNAMIC	Fa0/2

Switch#

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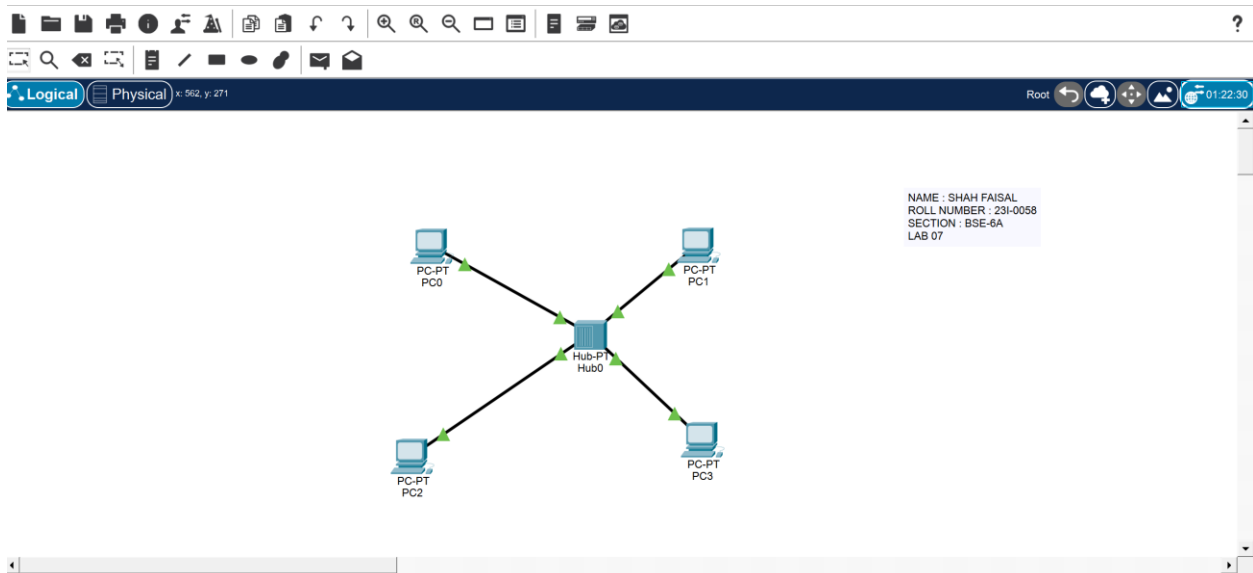
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Task 2: CSMA/CD Simulation (5 Marks)



PC1

Physical Config **Desktop** Programming Attributes

IP Configuration x

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::2D0:FFFF:FEA8:B2EE

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

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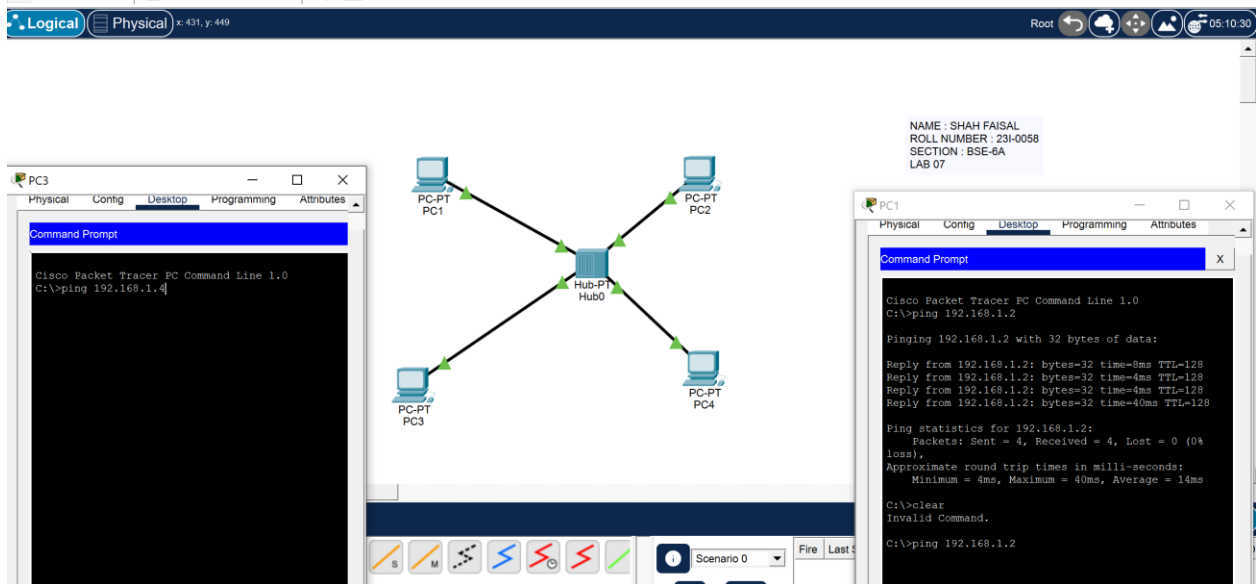
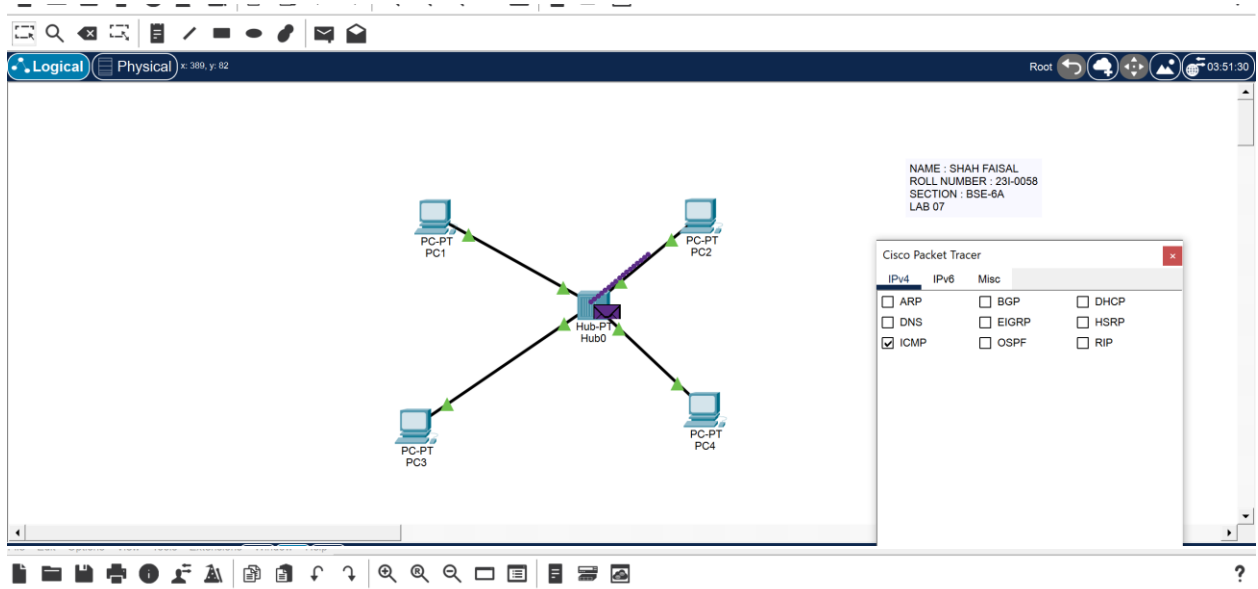
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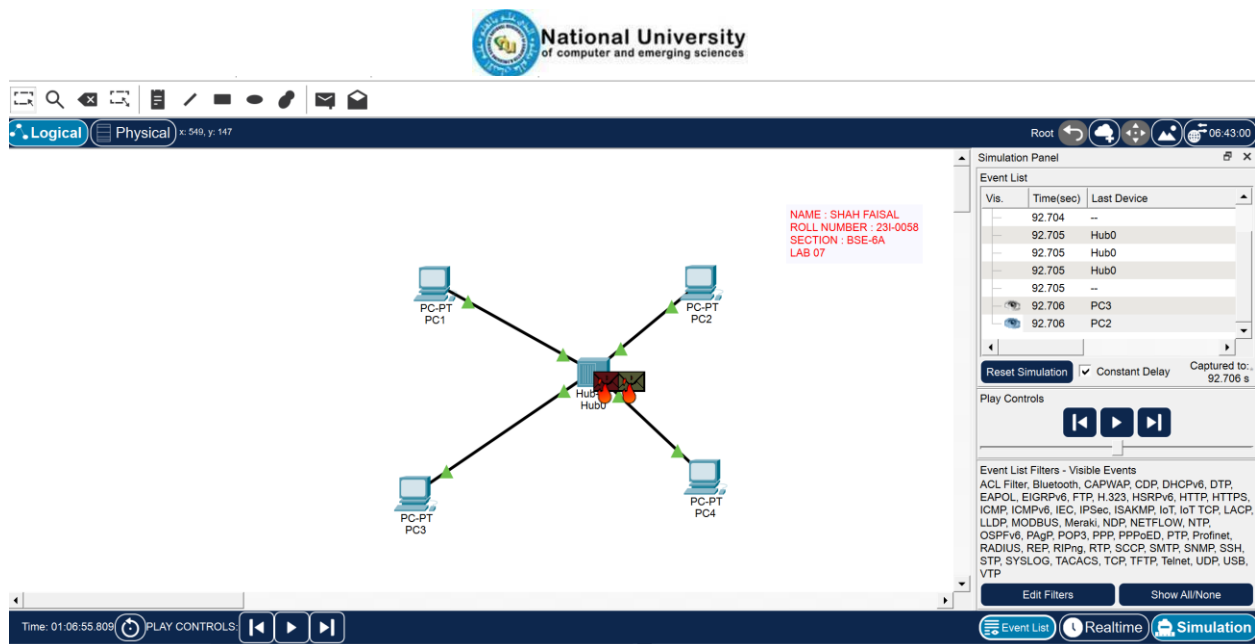
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How is collision handled in CSMA/CD?

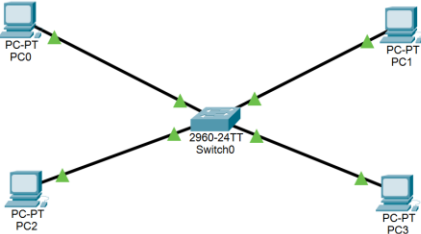
When a collision occurs in CSMA/CD:

1. **Detect:** Devices stop transmitting immediately upon detecting a collision
2. **Jam:** Each device sends a jam signal to notify all nodes
3. **Backoff:** Each device waits a random time (exponential backoff algorithm)
4. **Retry:** After waiting, devices listen again and retransmit when clear
5. **Give up:** After 16 failed attempts, the device stops trying

Task 3: Implementing VLAN (5 Marks)

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Logical Physical x: 570, y: 283 Root



```

graph TD
    Switch0[2960-24TT Switch0] --- PC0[PC-PT PC0]
    Switch0 --- PC1[PC-PT PC1]
    Switch0 --- PC2[PC-PT PC2]
    Switch0 --- PC3[PC-PT PC3]
  
```

Time: 00:06:27

Realtime Simu

Scenario 0

New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
------	-------------	--------	-------------	------	-------	-----------	----------	-----

Copper Straight-Through

IOS Command Line Interface

```
Switch(config)#interface fastEthernet1/1
%Invalid interface type and number
Switch(config)#interface fastEthernetinterface fastEthernet1/1
%Invalid interface type and number
Switch(config)#
Switch(config)#interface FastEthernet0/1
Switch(config-if)#enable
^
% Invalid input detected at '^' marker.

Switch(config-if)#configure terminal
^
% Invalid input detected at '^' marker.

Switch(config-if)#vlan 2
Switch(config-vlan)#name HR
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name Finance
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#end
Switch#show vlan brief
%SYS-5-CONFIG_I: Configured from console by console
```

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Switch0
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Physical Config **CLI** Attributes
Root

IOS Command Line Interface

```

Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name Finance
Switch(config-vlan)#exit
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/4
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#end
Switch#show vlan brief
%SYS-5-CONFIG_I: Configured from console by console
      
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24
2	HR	active	Gig0/1, Gig0/2
3	Finance	active	Fa0/1, Fa0/2
1002	fddi-default	active	Fa0/3, Fa0/4
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

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